

EXPERIMENT- 2

OBJECT:

To make house wiring including earthing for 1 Phase Energy Meter, M.C.B, Ceiling fan, tube light, three pin socket and a Lamp operated from two different positions.

APPARATUS REQUIRED:

S.NO.	ITEMS	RATING	QUANTITY
1.	Two way switch	6 ampere – 250 volt	2
2.	Switch	6 ampere – 250 volt	4
3.	Indicator	-----	1
4.	M.C.B.	6 ampere ,250 volt, DPST Type	1
5.	Holder	-----	2
6.	Bulb	100 watt , 250 volt	2
7.	Tester	100 watt , 250 volt	1
8.	3 Pin Socket	6 ampere , 250 volt	1
9.	Tube light	40 watt , 250 volt	1
10.	Fan with Reg.	12 Pole, Sweep 1200mm 220-230 voltage	1
11.	Energy Meter	1 Phase , 250 volt	1
12.	Wire Stripper	-----	1

THEORY:

A network of wires connecting various accessories for distribution of electrical energy from the supplier's meter board to the numerous electrical energy consuming devices such as lamps, fans and other domestic appliances through controlling and safety devices is known as wiring system. The supplier's service cable feeding an installation terminates in what is usually called the service fuses. In an ordinary house the service fuse is called as service cutout. Such cutouts including

service meters remain the property of the supplier and represent the furthest point of the supplier responsibility. The point at which the consumer's wiring is connected into cutout is known as point of commencement of supply or consumer's terminals. From consumer terminals onwards the supply cables are entirely under the control of consumer's and so laid out as per his selection. A typical house wiring circuit is shown in fig. 2.1

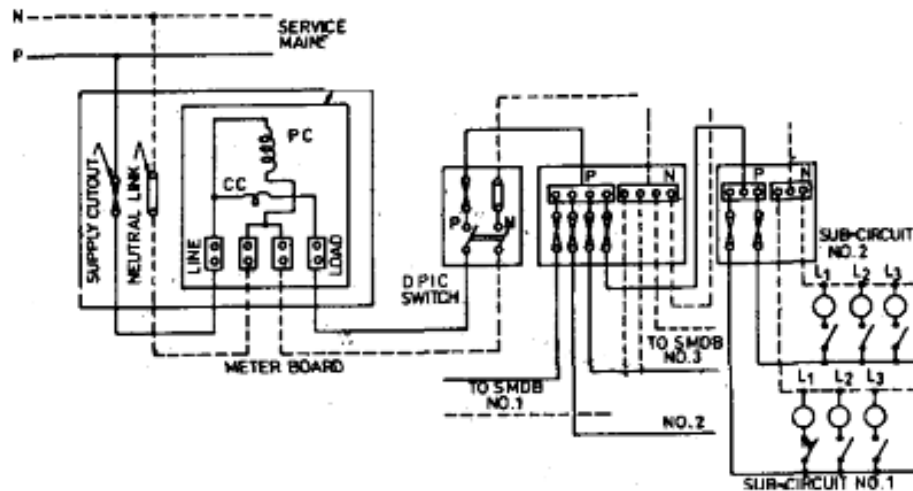


Fig 2.1 Connection diagram of House wiring

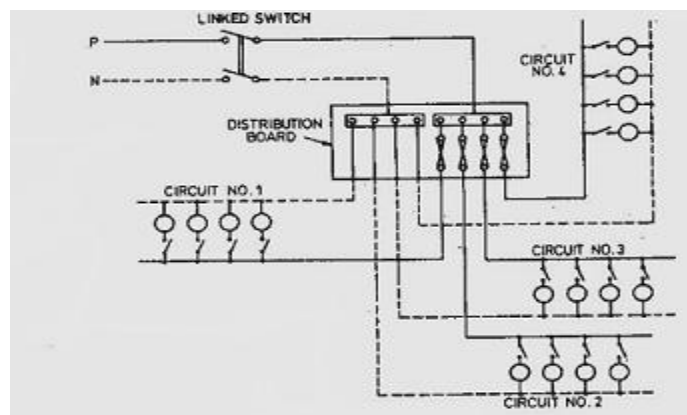


Fig 2.2 Distribution Board System

In distribution board system, which is most commonly adopted for distribution of electrical energy in a building, the fuses of various circuits are grouped together on a distribution board, sometimes

simply known as fuse board. The two copper strips, known as bus-bars, fixed in a distribution board of hard wood or metal case are connected to the supply main through a linked switch so that the installation can be switched off as a whole from both the poles of supply if required. A fuse is inserted in the +ve or phase pole of each circuit so that each circuit is connected up through its own particular fuse.

In large buildings, however, if only one distribution board were used, some of the points would be at a considerable distance from it and in such cases it is advisable to employ sub-distribution boards either to save cable or to prevent too great voltage drop at the more distant points (lamps or fans or other appliances). In such cases main distribution board controls the circuit to each sub-distribution board from which the sub-circuits are taken, as shown in fig. (a)

The number of circuits and sub-circuits are decided as per number of points to be wired and load to be connected to the supply system. For determination of load of an installation the following ratings may be assumed unless the values are known or specified.

- a) Fluorescent lamps — 40 watts.
- b) Incandescent lamps, fans, and socket outlets — 60 watts.
- c) Power socket-outlets — 1,000 watts.
- d) Exhaust fans — as per capacity of exhaust fans.

There are number of methods of installing a wire system.

- Cleat wiring
- Casing Wiring
- T.R.S. Wiring
- Metal Sheathed Wiring
- Conduit Wiring
- PVC Conduit Wiring

Generally, for wiring in the house 20 SWG wire is used and for Earthing 14 SWG is used.

HOUSE WIRE LAYOUT- Two wire from RSEB Pole bring 230 volt A.C. to our house. Our house is Phase (P) and other is neutral.

ENERGY METER- It may be disk type, Conduction Meter or Electronic Meter.

SOCKET- Socket has three types of terminals-

- Neutral,
- Phase ,

- Earthing

Earthing is of big size and other two are same.

M.C.B.(Miniature Circuit Breaker) normally, in House wiring, 2 Pole M.C.B. is used. As we know the fuse and M.C.B. are in fact is used to isolate a system from supply in case of:-

- Over load
- Short circuit

EARTHING:-

We know that Earthing is provided for the safety of both, Human Beings and Equipments. It is of two types-

- Plate earthing
- Pipe earthing

FACTOR AFFECTING EARTH RESISTANCE:-

The earth electrode resistance depends on electrical resistivity of soil which in turn depends upon-

- Nature of soil
- Extent of moisture
- Presence of suitable salt in moisture

EARTH ELECTRODE

It is of two types-

- rod and pipe electrode
- plate electrode

CIRCUIT DIAGRAM:

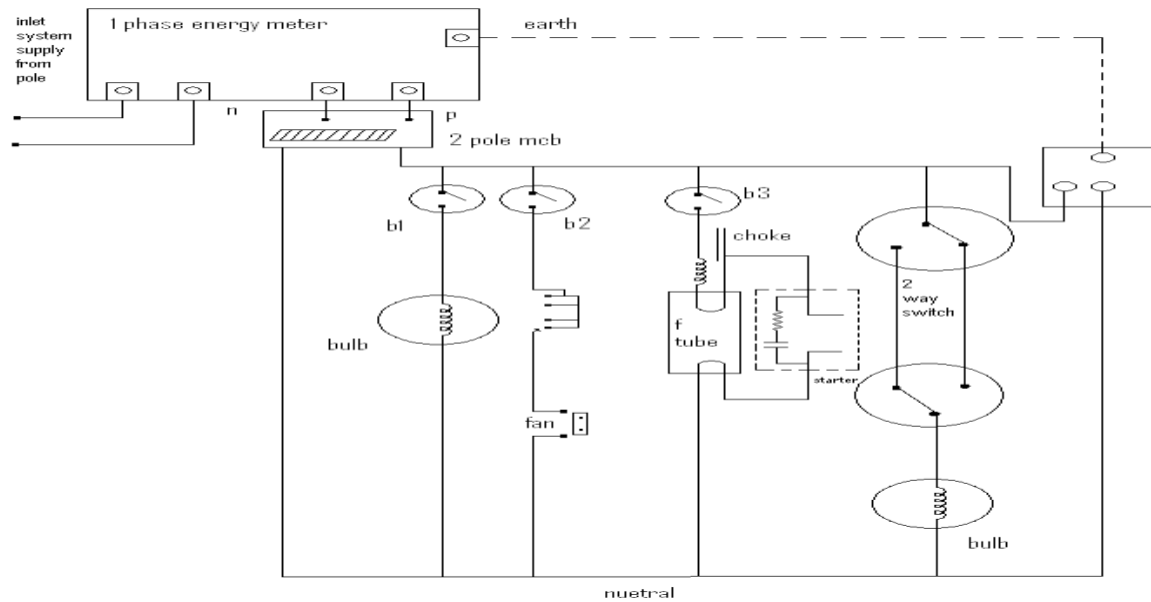


Fig 4.3 Circuit Diagram of House Wiring with Earthing

RESULT:

We make connection and get result as shown in Observation Table.

PRECAUTIONS:

1. Connections should be right and tight.
2. Do not touch any wire.
3. Do not keep any joint open.
4. Use M.C.B. and switches of proper current rating.

VIVA VOICE

1. What kind of wires used in house wiring? State size/gauge
2. What is the difference between one way and two way switch?
3. What is the specification of MCB used in house wiring? Also discuss about MCB Box.
4. What is the specification of smooth fan regulators used in ceiling fans?